

Lab: Little Man Computer (LMC)

IMPORTANT: Do NOT submit written answers for this lab. The questions below are provided strictly as a **self-study guide** to help you understand the concepts. Your final grade for this lab will be determined entirely by the automated Moodle Quiz, which you should take only *after* completing these exercises.

Background: Please access the emulator page: <https://www.peterhigginson.co.uk/LMC>

This website is an online emulator developed by Peter Higginson. It is a pedagogical proxy of a basic von Neumann Architecture central processing unit, with one channel for input and another for output, with no external storage. It has been a favorite tool to teach assembler programming since it was developed by Dr. Stuart Madnick in 1965.

Self-Study Activity 1: The "Add" Program

1. Click on the drop-down menu (Options) and select "add".
2. Run the program.
 - *Self-Check Q:* What does each instruction line do?
 - *Self-Check Q:* What is the overall purpose of this program?

Self-Study Activity 2: Instruction Tracing Look at the handwritten program table provided in your lecture notes.

- *Self-Check Q:* Notice the numeric codes (opcodes) next to the mnemonics. For example, LDA is (5), ADD is (1), and STO is (3). How does the CPU know the difference between an instruction and plain data?
- *Self-Check Q:* Follow the math in the "RESULT" column. If LDA loads 001 from mailbox 12, and ADD adds 390 from mailbox 15, what is the final stored result?

Self-Study Activity 3: Bubble Sort

1. From the dropdown menu, select "bubble sort".
 - *Self-Check Q:* Describe what the program does. Does it have ANY connection to the practice Linux shell scripts seen in lab 5? (Hint: Think about loops and conditional logic).

Self-Study Activity 4: Max Find (Custom Code) Type the following code into the **Assembly Language Code** box and run it with the inputs 15 and 25:

INP

STO 90

INP

STO 91

SUB 90

BRP 10

LDA 90

BRA 11

LDA 91

OUT

HLT

- *Self-Check Q:* How does subtracting one number from another (using **SUB**) help the computer determine which number is larger?
- *Self-Check Q:* What happens when the **BRP** (Branch if Positive) condition is met versus when it is not met?